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# AEGIS: A Comparative Economic Incentive-Design Approach to Breach Accountability and Cyber-Insurance for Fintech and E-Commerce, Evidence from Bangladesh for the Global South

## 1. Aims and Central Question

When personal data leaks, the firm that held it and the people it describes rarely bear the same cost. This proposal asks how to narrow that gap in Bangladesh's fintech and e-commerce sector, and how much of the answer could be reused elsewhere in the Global South. Its output is a costed, testable design called AEGIS, built from five instruments: **A**ccountability (a fixed deadline for telling affected customers), **E**nforcement (penalties scaled to the harm done, not only to company turnover), **G**overnance (named board-level responsibility), **I**nsurance (premium discounts linked to verified security controls) and **S**tandards (a minimum control set that regulators and insurers can both check).

**Central question:** Which mix of notification duties, enforceable liability and insurance-linked pricing gives Bangladeshi fintech and e-commerce firms a stronger economic reason to protect customer data, and which parts of that mix would carry over to other Global South markets?

## 2. Introduction: Problem, Evidence and Gap

Bangladesh now settles a large share of everyday payments on private platforms. IMF survey data record 238.7 million registered mobile-money accounts in 2024, of which 88.9 million were active, and 7.29 billion transactions worth BDT 17.37 trillion, or 34.7% of GDP, with 1.83 million registered agent outlets [1]. Even the active count shows how much personal and financial data sits inside private firms and their agent networks.

Recent incidents show where the incentive problem lies. In March 2026 Shwapno, the country's largest supermarket chain, said attackers were demanding US$1.5 million (over Tk 18.3 crore) after the names, phone numbers and purchase histories of over 40 lakh registered customers surfaced online; it had issued no public warning to them [2]. A general diary it lodged on 28 March dates the ransomware attack to 19 August 2025 and the leak of over 410 GB to 17 March 2026, and its administration head said the technical team had earlier assured management that the data was recovered [3]. The managing director instead told *The Daily Star* that the intrusion began in December 2025 [2], while a later commentary reads the case as seven months of silence [4]. The start date thus differs by about four months, which matters for measuring notice delay (Step 1); and the customers who carried most of the risk had no part in the assurance management relied on.

The legal position is unsettled. The Personal Data Protection Ordinance (No. 61 of 2025, 6 November 2025) was replaced by the Personal Data Protection Act, 2026 (No. 63 of 2026, 10 April 2026) [5]. A *Daily Star* summary of the draft Ordinance said breaches must be reported to the national data authority [6], whereas a July 2026 commentary says the Act has no mandatory breach-notification requirement [4]. On the reading adopted here, each is partly right. Section 20, identical in both texts, obliges a controller to inform the authority when a breach is likely to cause a data subject significant harm, within a period that regulations are to fix, and says nothing about telling the individuals [5]. Sections 31–35 (complaints, fines, compensation) start no earlier than 18 months after promulgation, and the fine for a security failure is capped at Tk 25 lakh, about one seventy-third of the Shwapno ransom demand [5]. The enforcer sits inside the executive [4], [7], and the August 2025 intrusion predates the Ordinance, so whether the silence was a legal failure or a legal gap turns on dates and regulations, which Step 1 checks.

The problem is wider than Bangladesh. The IMF reports a rising risk of extreme direct losses to firms (at least USD 2.5 billion), indirect losses that tend to exceed reported direct ones, and inadequate cyber policy frameworks in many emerging economies [8]; the World Economic Forum reports that 31% of respondents have low confidence in their nation's ability to respond to a major incident, up from 26% [9]. The best-known cost benchmark, US$4.44 million per breach in 2025, counts costs to the breached organisation, not to the people whose data was taken [10].

Most of the tested evidence on disclosure laws, breach litigation and cyber-insurance pricing comes from the United States and Europe (Section 3). Few studies examine how these instruments work together where supporting institutions, such as an independent regulator and shared loss data, are still being built. AEGIS is designed for that setting.

## 3. Literature Review

**Incentives.** Anderson and Moore argue that incentives matter as much as technical design, because the people who could prevent a failure often do not bear its cost [11]. Gordon and Loeb give this a formal shape: a firm should spend on protection only a fraction of its own expected loss, no more than about 37% under their assumptions [12]. Gordon and colleagues extend the model to externalities and show how private spending falls short of the efficient level [13]. The implication here is direct: if customers carry most of the loss, the firm's own expected loss is small and its rational spending ceiling sits below what the wider economy would want. Woods and Böhme add that cyber losses are not exceptional in size and that stock-market reactions to incidents have become less damaging over time [14], so market punishment alone is a weak substitute for a rule.

**Disclosure and liability.** Romanosky, Telang and Acquisti report that US breach-disclosure laws cut identity theft from breaches by about 6% on average [15]. In a study of more than 230 federal breach lawsuits, Romanosky, Hoffman and Acquisti found that a firm's odds of being sued were 3.5 times higher when individuals suffered financial harm and six times lower when the firm offered free credit monitoring [16]. Both findings rely on courts that work at scale, which limits how far they transfer. Penalty size is not the whole story either: liability cannot deter beyond what a firm can pay [17], and expected punishment is the fine multiplied by the chance of being caught [18], so a low cap and weak detection compound each other.

**Insurance.** Romanosky and colleagues analysed real cyber-insurance policies and note that little practical information on policy content and pricing is public, even though carriers are often seen as holding the best information about cyber risk [19]. Where filings are public, as in the United States, loss distributions can be inferred from them [20]. The harder problem is verification: premiums track security posture that is often self-reported, so insurers audit after claims and discounts invite misreporting [21]. Insurance can carry a security signal only if the controls it prices can be observed and checked, which is the design problem behind the "I" and "S" in AEGIS. Skeoch's extension of the Gordon–Loeb model to insurance [22] is the modelling starting point for Step 5.

**Context and capacity.** Madnick traces a 20% rise in data breaches from 2022 to 2023 to cloud misconfiguration, new ransomware tactics and exploited vendor systems [23], a route that resembles the "shadow copies" described in the *Daily Star* commentary, where 174 organisations connect to Bangladesh's national ID database and keep their own copies of what they retrieve [4]. The one peer-reviewed reading of the new Bangladeshi regime located for this proposal finds regulators embedded in executive authority and enforcement that assumes capacity not yet in place [7]. The Oxford Global Cyber Security Capacity Centre's maturity model offers a five-dimension frame for comparing national capacity [24], which this study uses to judge transferability.

**Gap and implications for design.** Three points follow. First, the instruments have mostly been tested one at a time. Second, several depend on courts or insurer data that may be thin in a market like Bangladesh's. Third, Bangladeshi material so far describes incidents and statutes without linking them to a firm's spending decision. A design for such a market should therefore lean on triggers that can be checked without a trial, such as the date customers were told or a control an auditor can confirm.

## 4. Research Questions

Each question is tied to a method step in Section 5.

**RQ1.** How large is the gap between what a breach costs a Bangladeshi fintech or e-commerce firm and what it costs its customers, judged by three observable proxies? (a) time from intrusion to customer notice; (b) remedy offered to customers; (c) legal or regulatory consequence for the firm. *[Step 1]*

**RQ2.** How do Bangladesh's notification, liability and governance rules compare with those of the EU, Kenya and India on the features that create an economic incentive? (a) notification deadline and audience; (b) penalty base and route to compensation; (c) regulator independence. *[Step 2]*

**RQ3.** Would firms change their security spending under insurance-linked pricing, and under which rule combinations? (a) stated preference between rule bundles; (b) insurers' access to usable loss data; (c) modelled effect on the firm's optimal spending. *[Steps 3 to 5]*

**RQ4.** Which AEGIS instruments would another Global South market need to adapt, and what preconditions do they need? (a) capacity gaps against the Oxford model; (b) expert agreement on each instrument. *[Steps 2 and 5]*

## 5. Methodology

The design is mixed-methods, in five steps ordered so that each supplies inputs to the next.

**Step 1: Incident and legal-text mapping (RQ1).** Build a register of publicly documented Bangladeshi fintech, e-commerce and related data incidents from 2023 to 2026, using press reports, police filings, regulator statements and investigative reports such as Dismislab's, each traced to its earliest original report. The target is at least 20 incidents; if fewer qualify, the count is reported as found. Each is coded for sector, data type, records exposed, dates of intrusion, detection and notice, remedy, legal consequence, and whether the Act applied at the time. Where sources conflict, as with Shwapno, the register keeps the date range and reports notice delay at both bounds. A second coder codes all incidents on dates and remedy and a random 20% on other fields; agreement is reported as Cohen's kappa and, for dates, median absolute difference. No leaked file is downloaded, bought or opened. The same step codes the Act in force, not the Ordinance it replaced, and checks whether regulations under section 20 have fixed a notification period.

**Step 2: Comparative policy analysis (RQ2, RQ4).** Compare Bangladesh with the EU's GDPR [25], Kenya [26] and India [27], chosen for their spread in income level and regulator design. Five features (notification deadline and audience, penalty base, route to compensation, regulator independence, and any link to insurance or technical standards) are scored 0 to 2 from the statutory text by two coders working separately, with weighted kappa and disagreements settled by discussion. Each score is tagged in force or not yet enforceable, since Bangladesh defers its fines. A first reading already shows a wide spread: 72 hours for notice to the regulator in the GDPR and Kenya, no statutory deadline in Bangladesh, and maximum fines for a security failure from Tk 25 lakh to ₹250 crore in India. Results are read against the five dimensions of the Oxford capacity model [24].

**Step 3: Firm survey with choice tasks (RQ3).** Survey 120 firms: 40 fintech and payment providers, 50 e-commerce sellers and platforms, and 30 other digital SMEs, recruited through business associations, university networks and snowball referral (a purposive sample). Each respondent completes four choice tasks drawn from a D-efficient design [28]. In each, two hypothetical rule bundles differ in notification deadline (7, 30 or 90 days), penalty basis (fixed sum or share of turnover) and premium discount for verified controls (0%, 10% or 25%), and the respondent picks the bundle that would push them harder to raise security spending, or answers that neither would. That yields 480 choices, analysed with a conditional logit model with standard errors clustered by respondent and reported as odds ratios with 95% confidence intervals. A common rule of thumb for main effects, $n \geq 500c/(ta)$ [29], asks for about 190 respondents here ($c = 3$, $t = 4$, $a = 2$), so results from 120 firms are exploratory and no subgroup effects are claimed.

**Step 4: Expert interviews (RQ3, RQ4).** Twelve semi-structured interviews: three regulators or policy officials, three insurers or brokers, three fintech or e-commerce security leads, and three academics or consumer advocates. Topics are insurer data availability (whether local insurers underwrite standalone cyber cover, and whether tariffs are open to a modeller, as US filings are [20]), enforcement capacity and the cost of each instrument. With four groups of three, findings are reported by group as expert elicitation, not saturated themes. Transcripts are coded by thematic analysis [30].

**Step 5: Modelling and validation (RQ3, RQ4).** Extend the Gordon–Loeb model [12] with (i) customer loss borne outside the firm, following Gordon and colleagues [13], (ii) an expected penalty and notification cost, capped at what the firm can pay [17], and (iii) a premium discount, building on Skeoch [22]. The 1/e ceiling is not assumed; more than one breach-probability function is run. Parameter ranges come from Steps 1, 3 and 4; a Monte Carlo run of 10,000 draws shows how optimal spending shifts under each instrument, with sensitivity results. The draft AEGIS blueprint then goes to a two-round Delphi panel of 10 to 12 experts, and an instrument is kept if at least 70% of panellists rate it feasible.

## 6. Expected Outcomes and Limitations

**Outcomes.** (1) A register of documented incidents with a measured notice delay and a firm-versus-customer comparison on three proxies (RQ1). (2) A scored comparison matrix for four jurisdictions, tagged in force or not yet enforceable (RQ2). (3) Odds ratios showing which rule bundles firms say would raise spending, and a calibrated model showing the direction and rough size of the effect (RQ3). (4) The AEGIS blueprint: five instruments, each with a verifiable trigger, a named enforcer and a note on preconditions for other Global South markets (RQ4).

**Limitations.** The survey sample is purposive and below the rule-of-thumb size, so results describe respondents, not all firms, and stated preference can differ from behaviour. The incident register depends on press coverage and will miss breaches never detected or reported, and the firm-versus-customer gap is judged through three proxies, not a monetary loss. Bangladesh's law is changing quickly (an Ordinance became an Act within five months), so the

analysis is fixed to the text in force when coding is done, and a Bengali-speaking legal practitioner checks the coding. The model shows direction and sensitivity, not forecasts. The study offers policy analysis, not legal advice.

## 7. Practicalities, Ethics and Post-Program Plan

**Timeline (12 weeks).** Weeks 1–2: incident register and Act text (Step 1). Weeks 3–4: comparative scoring (Step 2). Weeks 5–6: survey design, pilot with eight firms to test the wording, interview guide. Weeks 7–9: fieldwork for Steps 3 and 4. Week 10: modelling (Step 5). Weeks 11–12: Delphi rounds, blueprint and write-up.

**Ethics.** Participation is voluntary with recorded consent. Firms and interviewees are anonymised and firm names are withheld from results. Data are stored encrypted and deleted after the study period. The protocol will go to the institution's ethics committee before fieldwork. The study uses no leaked personal data.

**Post-program.** A short policy brief will be offered to the national data authority, the central bank's payment systems department and the e-commerce and fintech associations. Anonymised survey data and coding sheets will be shared. The survey can be repeated after twelve months to see whether stated preferences moved once enforcement began.

## 8. Conclusion

AEGIS starts from one observation: a firm protects what it expects to pay for. In Bangladesh the gap between the firm's cost and the customer's cost looks wide in recent cases, and the rules meant to close it are thin: on a first reading, notice goes only to the authority, on a timetable left to regulation. By coding real incidents, comparing four legal designs, testing firm preferences and validating the result with experts, the study will produce a blueprint whose every instrument can be checked without proving harm in court. Its value beyond Bangladesh is a method that other markets with young institutions can repeat with their own data.

## References


[1]International Monetary Fund, "Financial Access Survey: mobile money series for Bangladesh (series BGDFCMARNUM, BGDFCMAANUM, BGDFCMTVXDC and others)," FRED, Federal Reserve Bank of St. Louis. [Online]. Available: https://fred.stlouisfed.org/series/BGDFCMARNUM. Accessed: Sep. 21, 2026.

[2]S. Halder, "Shwapno's database hacked; 40 lakh customers' details at risk," *The Daily Star*, Mar. 28, 2026. [Online]. Available: https://www.thedailystar.net/business/news/shwapnos-database-hacked-40-lakh-customers-details-risk-4137566

[3]TBS Report, "Shwapno files GD seven months after customer data breach," *The Business Standard*, Mar. 29, 2026. [Online]. Available: https://www.tbsnews.net/bangladesh/shwapno-files-gd-seven-months-after-customer-data-breach-1396991

[4]M. A. Manab, "Why Bangladesh's new data protection law may fail to protect your data," *The Daily Star*, via Asia News Network, Jul. 8, 2026. [Online]. Available: https://asianews.network/why-bangladeshs-new-data-protection-law-may-fail-to-protect-your-data/

[5]Personal Data Protection Act, 2026 (Act No. 63 of 2026), Bangladesh, Apr. 10, 2026 (Bengali text), repealing the Personal Data Protection Ordinance, 2025 (Ordinance No. 61 of 2025), Nov. 6, 2025. [Online]. Available: https://bdlaws.minlaw.gov.bd/act-1692.html

[6]Z. Rabbi, "Bangladesh's Personal Data Protection Ordinance 2025: Key takeaways," *The Daily Star*, Oct. 21, 2025. [Online]. Available: https://www.thedailystar.net/tech-startup/news/bangladeshs-personal-data-protection-ordinance-2025-key-takeaways-4015401

[7]P. Saha, A. S. Chan, and S. Sultana, "When data protection fails to protect: Law, power, and postcolonial governance in Bangladesh," in *Proc. ACM SIGCAS/SIGCHI Conf. Comput. Sustain. Soc. (COMPASS)*, 2026, doi: 10.1145/3811242.3819089.

[8]International Monetary Fund, "Cyber risk: A growing concern for macrofinancial stability," in *Global Financial Stability Report: The Last Mile: Financial Vulnerabilities and Risks*, ch. 3, Washington, DC, USA, Apr. 2024.

[9]World Economic Forum, in collaboration with Accenture, *Global Cybersecurity Outlook 2026*, Geneva, Switzerland, Jan. 2026. [Online]. Available: https://www.weforum.org/publications/global-cybersecurity-outlook-2026/

[10]IBM Security and Ponemon Institute, *Cost of a Data Breach Report 2025*, Armonk, NY, USA, Jul. 2025.

[11]R. Anderson and T. Moore, "The economics of information security," *Science*, vol. 314, no. 5799, pp. 610–613, Oct. 2006, doi: 10.1126/science.1130992.

[12]L. A. Gordon and M. P. Loeb, "The economics of information security investment," *ACM Trans. Inf. Syst. Secur.*, vol. 5, no. 4, pp. 438–457, Nov. 2002, doi: 10.1145/581271.581274.

[13]L. A. Gordon, M. P. Loeb, W. Lucyshyn, and L. Zhou, "Externalities and the magnitude of cyber security underinvestment by private sector firms: A modification of the Gordon–Loeb model," *J. Inf. Secur.*, vol. 6, no. 1, pp. 24–30, Jan. 2015.

[14]D. W. Woods and R. Böhme, "SoK: Quantifying cyber risk," in *Proc. IEEE Symp. Security and Privacy (SP)*, 2021, doi: 10.1109/SP40001.2021.00053.

[15]S. Romanosky, R. Telang, and A. Acquisti, "Do data breach disclosure laws reduce identity theft?," *J. Policy Anal. Manage.*, vol. 30, no. 2, pp. 256–286, 2011.

[16]S. Romanosky, D. Hoffman, and A. Acquisti, "Empirical analysis of data breach litigation," *J. Empirical Legal Stud.*, vol. 11, no. 1, pp. 74–104, Mar. 2014, doi: 10.1111/jels.12035.

[17]S. Shavell, "The judgment proof problem," *Int. Rev. Law Econ.*, vol. 6, no. 1, pp. 45–58, 1986.

[18]G. S. Becker, "Crime and punishment: An economic approach," *J. Polit. Econ.*, vol. 76, no. 2, pp. 169–217, 1968.

[19]S. Romanosky, L. Ablon, A. Kuehn, and T. Jones, "Content analysis of cyber insurance policies: How do carriers price cyber risk?," *J. Cybersecurity*, vol. 5, no. 1, Art. no. tyz002, 2019, doi: 10.1093/cybsec/tyz002.

[20]D. W. Woods, T. Moore, and A. C. Simpson, "The county fair cyber loss distribution: Drawing inferences from insurance prices," *Digit. Threats: Res. Pract.*, vol. 2, no. 2, Art. no. 10, 2021, doi: 10.1145/3434403.

[21]S. Panda, D. W. Woods, A. Laszka, A. Fielder, and E. Panaousis, "Post-incident audits on cyber insurance discounts," *Comput. Secur.*, vol. 87, Art. no. 101593, 2019, doi: 10.1016/j.cose.2019.101593.

[22]R. K. Skeoch, "Expanding the Gordon–Loeb model to cyber-insurance," *Comput. Secur.*, Art. no. 102533, 2021, doi: 10.1016/j.cose.2021.102533.

[23]S. Madnick, "Why data breaches spiked in 2023," *Harvard Business Review*, Feb. 19, 2024. [Online]. Available: https://hbr.org/2024/02/why-data-breaches-spiked-in-2023

[24] Global Cyber Security Capacity Centre, Univ. of Oxford, *Cybersecurity Capacity Maturity Model for Nations (CMM), 2021 Edition*, Mar. 2021, doi: 10.2139/ssrn.3822153.

[25] Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 (General Data Protection Regulation), *Official J. Eur. Union*, L 119, May 4, 2016.

[26] Data Protection Act, No. 24 of 2019, Laws of Kenya.

[27] Digital Personal Data Protection Act, 2023 (No. 22 of 2023), India.

[28] F. R. Johnson *et al.*, "Constructing experimental designs for discrete-choice experiments: Report of the ISPOR Conjoint Analysis Experimental Design Good Research Practices Task Force," *Value Health*, vol. 16, no. 1, pp. 3–13, 2013.

[29] B. K. Orme, *Getting Started with Conjoint Analysis: Strategies for Product Design and Pricing Research*, 2nd ed. Madison, WI, USA: Research Publishers, 2010.

[30] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qual. Res. Psychol.*, vol. 3, no. 2, pp. 77–101, 2006.